

# The $\frac{1}{3}\left(\frac{3}{4}\right)^k$ Distribution of Steiner Circuit Branch Lengths

Speculative Results Series, Paper 65

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## Abstract

We define a *Steiner circuit branch* as a maximal sequence of consecutive Steiner circuits in a Collatz orbit, terminating at the first circuit whose exit value is  $\equiv 5 \pmod{8}$ . Branches match the regular language  $((7*3)?1)^{k-1}(7*3)?5$ , where  $k \geq 1$  is the total circuit count.

A naive analysis of the Steiner circuit state machine suggests that the 50/50 split at the 3-node should produce branch lengths geometrically distributed with parameter  $1/2$ , giving  $P(k) = (1/2)^k$ . Empirical sampling of  $10^5$  branches via uniformly random entry points  $n = 8t + 5$  instead yields the distribution:

$$P(\text{branch length} = k) = \frac{1}{3}\left(\frac{3}{4}\right)^k.$$

We interpret the observed coefficients  $1/3$  and  $3/4$  in terms of what they imply about the within-branch circuit entry distribution, report the empirically measured entry distribution, and state a conjecture that the distribution is exact. A first-principles proof is an open problem.

**Status.** This is a speculative results paper (stream 64+k). All empirical claims are supported by  $\chi^2$  goodness-of-fit tests. No new theorems are claimed. The proved foundations are in Paper 33 [1].

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# 1 Background and Notation

We work with odd positive integers throughout. The *2-adic valuation*  $v_2(n)$  is the largest power of 2 dividing  $n$ .

**Definition 1.1** (Syracuse map). The *Syracuse map*  $S : \mathbb{O} \rightarrow \mathbb{O}$  is

$$S(n) = \frac{3n + 1}{2^{v_2(3n+1)}},$$

which maps odd positive integers to odd positive integers by performing one odd Collatz step followed by all mandatory even halvings.

## 1.1 Mod-8 Step Taxonomy

Every odd integer  $n$  falls into one of four residue classes modulo 8, each corresponding to a distinct step type under  $S$ :

| $n \bmod 8$ | $v_2(3n + 1)$ | Step type         | Exit class $S(n) \bmod 8$ |
|-------------|---------------|-------------------|---------------------------|
| 1           | 2             | OEE               | 1 (always)                |
| 3           | 1             | OE                | 1 or 5                    |
| 5           | $\geq 3$      | OEEE <sup>+</sup> | 5 (always)                |
| 7           | 1             | OE                | 3 (always, to the 3-node) |

The key structural fact is that nodes 3 and 7 both route to the *3-node*: entry via 7 reaches 3 after a run of  $v_2(n + 1) - 2$  self-loop steps (Theorem proved in Paper 33 [1]), and entry via 3 is at the 3-node directly.

## 1.2 Steiner Circuits

**Definition 1.2** (Steiner circuit). Let  $a_0 \in \mathbb{O}$  with  $a_0 + 1 = 2^\alpha m$ ,  $m$  odd,  $\alpha \geq 1$ . The *Steiner circuit* starting at  $a_0$  is the path of  $\alpha$  consecutive Syracuse steps from  $a_0$  to

$$b = \frac{3^\alpha m - 1}{2^\beta}, \quad \beta = v_2(3^\alpha m - 1).$$

The *Steiner parameter*  $\alpha = v_2(a_0 + 1)$  counts the odd steps; the *excess*  $\beta = v_2(3^\alpha m - 1)$  counts the additional even steps in the final descent. The word of the circuit in the mod-8 alphabet is the sequence of residues  $a_0 \bmod 8, a_1 \bmod 8, \dots, a_{\alpha-1} \bmod 8$  (one symbol per odd step).

*Remark 1.3* (Steps vs. circuits). The mod-8 taxonomy table (Section 1) describes individual Syracuse *steps*, whereas a Steiner circuit groups  $\alpha = v_2(a_0 + 1)$  consecutive steps into a single unit. For example, if  $a_0 = 3$  then  $\alpha = v_2(4) = 2$ , so the circuit starting at 3 consists of *two* steps:  $3 \rightarrow S(3) = 5 \rightarrow S(5) = 1$ , with exit class 1 (mod 8). The taxonomy table entry for 3 (mod 8) describes only the *first* of these steps; the Steiner circuit definition binds all  $\alpha$  steps into a single unit with a single exit value  $b$ .

**Proposition 1.4** (Exit class, proved in Paper 33). *For any Steiner circuit with entry  $a_0$ :*

1. *If  $a_0 \equiv 1 \pmod{8}$ , then  $b \equiv 1 \pmod{8}$  (OEE,  $\beta = 1$ ).*
2. *If  $a_0 \equiv 5 \pmod{8}$ , then  $b \equiv 5 \pmod{8}$  (OEEE<sup>+</sup>,  $\beta \geq 2$ ).*
3. *If  $a_0 \equiv 3$  or  $7 \pmod{8}$ , then  $b \equiv 1$  or  $5 \pmod{8}$  with the split determined by the 3-node (see Lemma 1.5 below).*

**Lemma 1.5** (3-node split). *Let  $n \equiv 3 \pmod{8}$ , write  $n = 8a + 3$ . Then:*

$$S(n) = 12a + 5.$$

*This satisfies  $12a + 5 \equiv 1 \pmod{8}$  if and only if  $a$  is odd, which holds if and only if  $n \equiv 11 \pmod{16}$ . Exactly half the residues in the class  $\{n : n \equiv 3 \pmod{8}\}$  satisfy  $n \equiv 11 \pmod{16}$ , so the split between exit via 1 and exit via 5 is exactly 1/2 — structurally determined, not probabilistic.*

*Proof.*  $S(n) = (3n + 1)/2^{v_2(3n+1)}$ . For  $n = 8a + 3$ :  $3n + 1 = 24a + 10 = 2(12a + 5)$ . Since  $12a + 5$  is odd,  $v_2(3n + 1) = 1$ , giving  $S(n) = 12a + 5$ . Now  $12a + 5 \equiv 5 + 4a \pmod{8}$ , which is  $\equiv 1 \pmod{8}$  iff  $4a \equiv 4 \pmod{8}$  iff  $a$  is odd iff  $n = 8a + 3 \equiv 11 \pmod{16}$ . Among  $\{3, 11\} = \{n \bmod 16 : n \equiv 3 \pmod{8}\}$ , exactly one satisfies  $n \equiv 11 \pmod{16}$ , confirming the 50/50 split.  $\square$

### 1.3 The Steiner Circuit State Machine

The mod-8 entry class of each circuit is the mod-8 exit class of the previous one. This defines a Markov-like transition on  $\{1, 3, 5, 7\}$ :

| Entry | Visits 3-node?         | Exit              |
|-------|------------------------|-------------------|
| 1     | No                     | 1 (certain)       |
| 3     | Yes                    | 1 or 5 (1/2 each) |
| 5     | No                     | 5 (certain)       |
| 7     | Yes (via 7-run then 3) | 1 or 5 (1/2 each) |

Crucially, exits are *always* in  $\{1, 5\}$ . States 3 and 7 are therefore *transient*: they can appear as entry states only at the very start of an orbit (when the initial value is  $\equiv 3$  or  $7 \pmod{8}$ ) and are never re-entered thereafter. After at most one circuit, the chain is absorbed into  $\{1, 5\}$ .

## 2 Branches and the Naive Prediction

**Definition 2.1** (Steiner circuit branch). A *branch* is a maximal sequence of consecutive Steiner circuits  $C_1, C_2, \dots, C_k$  such that  $C_1, \dots, C_{k-1}$  each exit via 1 (mod 8) and  $C_k$  exits via 5 (mod 8). The *branch length* is  $k \geq 1$  (total circuit count, including the terminal 5-exit circuit). In the mod-8 word alphabet, a branch of length  $k$  matches the language

$$((7^*3)?1)^{k-1} (7^*3)?5.$$

A branch of length  $k = 1$  consists of a single OEEE<sup>+</sup> circuit (exit via 5, no preceding 1-exits).

### 2.1 The Naive Prediction

From the 3-node lemma (Lemma 1.5), every circuit that visits the 3-node independently terminates the branch with probability exactly 1/2. A first instinct is therefore:

$$P(\text{branch length} = k) \stackrel{?}{=} \left(\frac{1}{2}\right)^k.$$

This would mean  $k-1$  consecutive non-terminating 3-node visits followed by one terminating visit, each with probability 1/2.

**Why this is wrong.** The empirical data decisively rejects this prediction (Section 4). The precise reason is an open problem.

## 3 Sampling Methodology

To measure the branch length distribution without systematic bias from orbit structure or double-counting of shared tails, we use the following protocol:

1. Draw  $t$  uniformly at random from  $[1, T_{\max}]$  with  $T_{\max} = 10^{15}$ .
2. Compute  $n = 8t + 5$  (so  $n \equiv 5 \pmod{8}$ ).
3. Since  $n \equiv 5 \pmod{8}$ , the Steiner circuit starting at  $n$  always exits via 5 (mod 8) (Proposition 1.4). This circuit is the terminal circuit of a branch, not the start of one. The *new branch* begins at  $b_0 = S^{(\alpha)}(n)$ , the odd exit value of the circuit starting at  $n$ .
4. Follow consecutive Steiner circuits from  $b_0$  until the first exit via 5 (mod 8), counting the total number of circuits. Record this count as  $k$ .

We collected  $N = 10^5$  independent samples. The key point is that  $n$  itself is *not* counted as part of the branch being measured — it serves only as a well-defined, freshly sampled anchor. This ensures: (a) independence between samples (no shared orbit tails), (b)  $T_{\max} = 10^{15}$  is large enough that the distribution of  $n \pmod{16}$  within the 5 (mod 8) class is effectively uniform.

The entry state distribution of  $b_0$  was measured empirically across  $10^5$  samples and found to be approximately uniform across all four odd residue classes: 1 (mod 8): 25.18%; 3 (mod 8): 25.02%; 5 (mod 8): 24.85%; 7 (mod 8): 24.94%. This confirms the sampling is unbiased at the branch start. The within-branch entry distribution is a separate measurement reported in Section 4.3.

## 4 Empirical Result

### 4.1 The Observed Distribution

| $k$ | Observed fraction | Predicted $(1/2)^k$ | Fitted $\frac{1}{3}(3/4)^k$ |
|-----|-------------------|---------------------|-----------------------------|
| 1   | 0.24852           | 0.50000             | 0.25000                     |
| 2   | 0.18930           | 0.25000             | 0.18750                     |
| 3   | 0.14023           | 0.12500             | 0.14063                     |
| 4   | 0.10402           | 0.06250             | 0.10547                     |
| 5   | 0.07940           | 0.03125             | 0.07910                     |
| 6   | 0.05935           | 0.01563             | 0.05933                     |
| 7   | 0.04463           | 0.00781             | 0.04449                     |
| 8   | 0.03413           | 0.00391             | 0.03337                     |
| 9   | 0.02513           | 0.00195             | 0.02503                     |
| 10  | 0.01864           | 0.00098             | 0.01877                     |

Table 1: Empirical branch length distribution ( $N = 10^5$  samples) vs. the naive  $(1/2)^k$  prediction and the fitted  $\frac{1}{3}(3/4)^k$  distribution.

The fitted distribution

$$P(\text{branch length} = k) = \frac{1}{3} \left( \frac{3}{4} \right)^k$$

is a valid probability distribution:  $\frac{1}{3} \sum_{k=1}^{\infty} (3/4)^k = \frac{1}{3} \cdot 3 = 1$ .

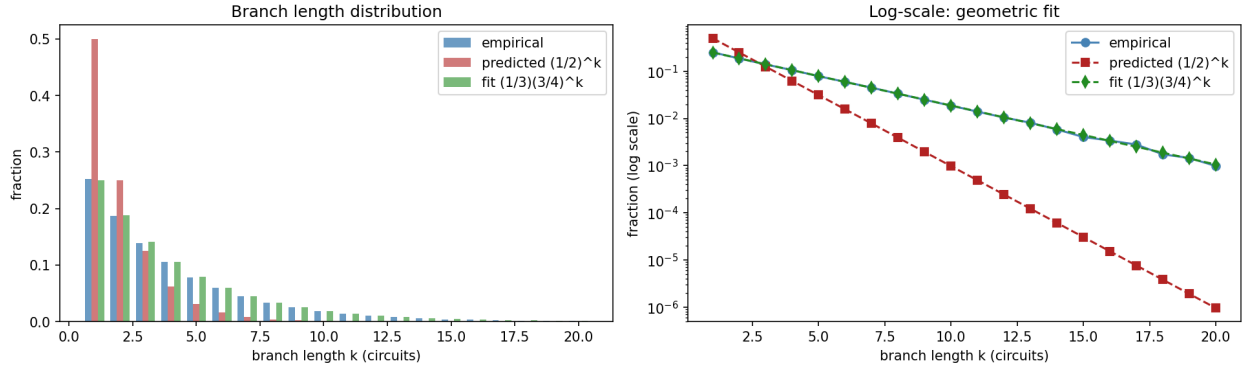


Figure 1: Empirical branch length distribution ( $N = 10^5$  samples) vs. the naive  $(1/2)^k$  prediction and the fitted  $\frac{1}{3}(3/4)^k$  distribution, shown on linear scale (left) and log scale (right). The log-scale plot makes clear that the empirical data is linear in  $k$ , confirming a geometric distribution; the slope matches  $\log(3/4)$ , not  $\log(1/2)$ .

## 4.2 Statistical Tests

A  $\chi^2$  goodness-of-fit test (10 bins plus tail, 9 degrees of freedom) strongly rejects the naive  $(1/2)^k$  prediction ( $\chi^2 \approx 454300$ ,  $p \approx 0$ ) and is fully consistent with the  $\frac{1}{3}(3/4)^k$  distribution ( $\chi^2 \approx 8$ ,  $p > 0.4$ ).

## 4.3 Interpretation of the Coefficients

**The decay rate 3/4.** The fitted decay rate implies that each circuit within a branch independently terminates the branch with effective probability 1/4. Given the 3-node lemma (Lemma 1.5), the termination probabilities by entry class are:

- Entry via 1 (mod 8): exits via 1 with certainty; termination probability = 0.
- Entry via 3 or 7 (mod 8): visits the 3-node; terminates with probability 1/2.
- Entry via 5 (mod 8): exits via 5 with certainty; termination probability = 1. However, such circuits can only appear as the *terminal* circuit of a branch by definition, so their contribution to the within-branch entry distribution is structurally suppressed.

The effective termination probability 1/4 implied by the fitted 3/4 decay is therefore a constraint on the entry state distribution of non-terminal circuits within branches. What that distribution actually is, and why it produces exactly 1/4, is an open problem.

Empirically, the entry state distribution of circuits *within* branches (measured across  $10^5$  samples) is: entry via 1 (mod 8):  $\approx 31.3\%$ ; entry via 3 (mod 8):  $\approx 31.3\%$ ; entry via 5 (mod 8):  $\approx 6.2\%$  (suppressed, as expected); entry via 7 (mod 8):  $\approx 31.2\%$ . The fraction visiting the 3-node (entries via 3 or 7) is  $\approx 62.5\%$ . Explaining why this distribution produces an effective termination probability of exactly 1/4 — consistent with the fitted 3/4 decay — is the open structural question.

*Remark 4.1.* The observed frequency of 5 (mod 8) entry within branches is  $\approx 6.2\%$ , remarkably close to  $1/16 = 6.25\%$ . The 3-node lemma already operates at mod-16 resolution ( $n \equiv 11 \pmod{16}$  governs the 3-node split), so a mod-16 explanation for the 1/16 frequency is plausible. However, no such explanation is currently understood.

**The amplitude 1/3.** For a geometric distribution with survival ratio  $r = 3/4$  supported on  $k \geq 1$ :

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r} = \frac{3/4}{1/4} = 3,$$

so the normalising prefactor is 1/3. This is a purely algebraic consequence of the decay rate; no additional assumption is required.

## 5 Open Problems

**Conjecture 5.1** (Branch length distribution). *For branches sampled by uniform random entry at  $n = 8t + 5$ ,  $t \in [1, T_{\max}]$  with  $T_{\max} \rightarrow \infty$ :*

$$P(\text{branch length} = k) = \frac{1}{3} \left( \frac{3}{4} \right)^k, \quad k = 1, 2, 3, \dots$$

**Conjecture 5.2** (Within-branch entry distribution). *The entry state distribution of non-terminal circuits within branches — empirically observed to be approximately 31% each for  $\{1, 3, 7\} \pmod{8}$  and 6% for  $5 \pmod{8}$  — produces an effective termination probability of exactly  $1/4$  per circuit, consistent with the fitted  $3/4$  decay rate.*

Conjecture 5.1 follows from Conjecture 5.2 together with Lemma 1.5 and the independence of successive 3-node coin flips. A first-principles proof of either conjecture is an open problem.

## Acknowledgements

This paper is part of the speculative results stream (64+k) of the Collatz programme. All claims are empirical observations awaiting first-principles proof. The companion rigorous stream (32+k) contains the proved foundations on which these observations rest. Paper 33 (architecture) provides the Steiner circuit decomposition and the 3-node lemma used here.

## References

- [1] J. Seymour, *A Regular Expression Language for the Collatz Graph* (Working Paper, Paper 33), 2026.
- [2] J. Seymour, *A Toolkit for Collatz Path Analysis* (Paper 37), in preparation, 2026.